Synthetic Data Statistical Comparison & Validation Report

Automated Generative AI Quality, Statistical Divergence & Privacy Assessment

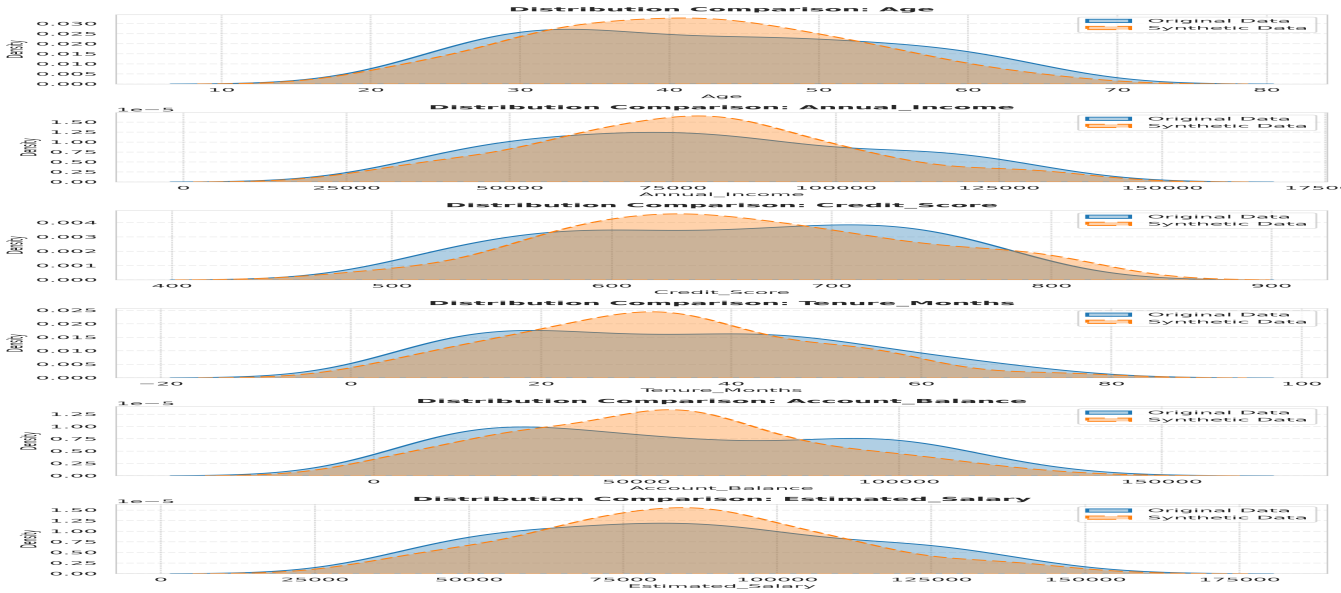
1. Executive Summary & Quality Scores

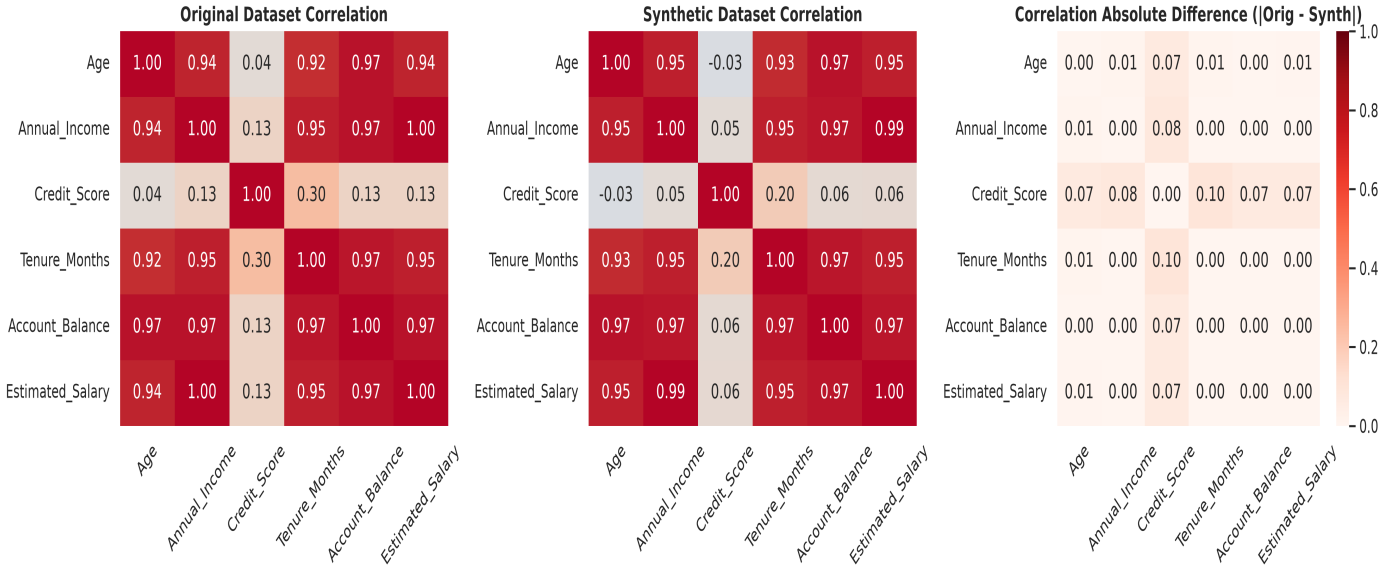
Metric	Score / Value	Assessment Benchmark
Overall Synthetic Data Quality Score	92.6%	>= 80% (High Quality)
Statistical Similarity Index	92.1%	Preservation of Marginal Distributions
Correlation Matrix Similarity	98.0%	Preservation of Inter-Feature Covariance
Privacy Protection Score (100 - Risk)	100.0%	>= 90% (Low Memorization Risk)
Median Distance to Closest Record (DCR)	0.6886	Higher indicates greater privacy protection
Exact Match Clones Count	0	0 is Optimal (Zero direct memorization)

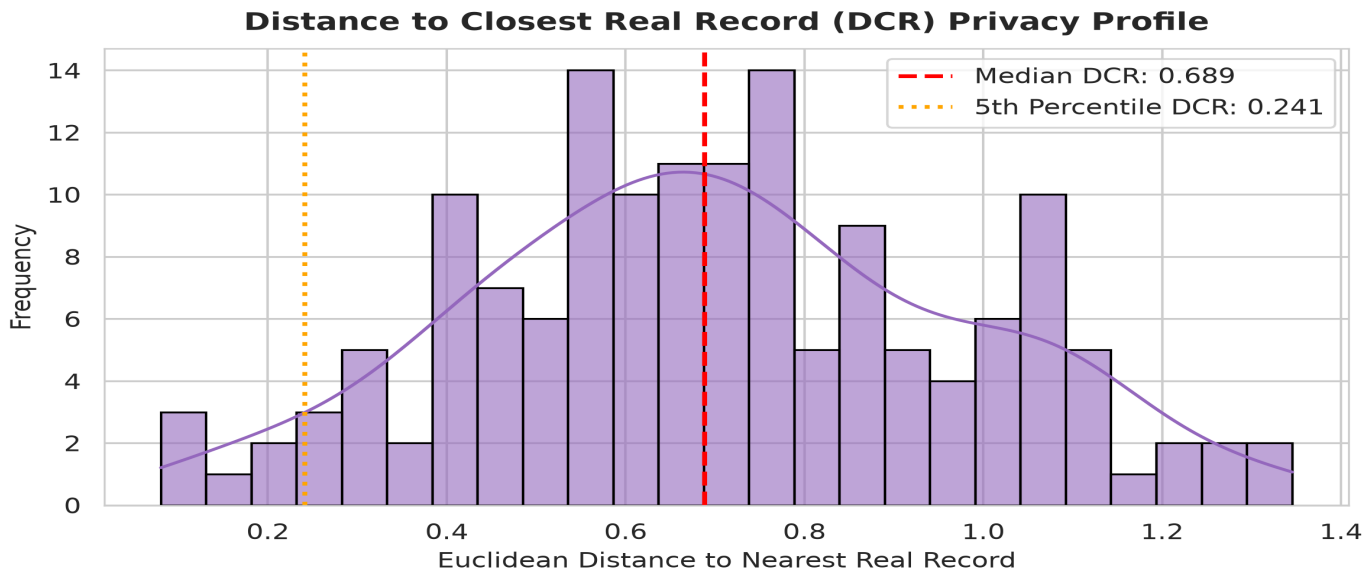
2. Numerical Features Kolmogorov-Smirnov (KS) Validation

Feature Name	KS Statistic (D)	p-value	Fidelity Score (1 - D)	Distribution Alignment
Age	0.1267	0.5647	87.3%	Strong Match
Annual_Income	0.1067	0.7692	89.3%	Strong Match
Credit_Score	0.0867	0.9311	91.3%	Strong Match
Tenure_Months	0.1267	0.5647	87.3%	Strong Match
Account_Balance	0.1600	0.2783	84.0%	Strong Match
Estimated_Salary	0.1133	0.7021	88.7%	Strong Match

3. Visual Distribution & Correlation Analysis


Distribution Comparison: Age
Distribution Comparison: Annual_Income
Distribution Comparison: Credit_Score
Distribution Comparison: Tenure_Months
Distribution Comparison: Account_Balance
Distribution Comparison: Estimated_Salary
Original Data
Synthetic Data
Density


Original Dataset Correlation
Synthetic Dataset Correlation
Correlation Absolute Difference (|Orig - Synth|)
Age
Annual_Income
Credit_Score
Tenure_Months
Account_Balance
Estimated_Salary


Distance to Closest Real Record (DCR) Privacy Profile
Median DCR: 0.689
5th Percentile DCR: 0.241
Frequency
Euclidean Distance to Nearest Real Record